

SMART INDIA HACKATHON 2026 | SIH26167 | DEPARTMENT OF SPACE, ISRO

SatQuery AI

An Agentic, Multi-Tier Vision-Language System for
Conversational Satellite Image Understanding

Technical Report: Architecture, LoRA Fine-Tuning
Methodology, and Empirical Evaluation

Base model Qwen2.5-VL-3B-Instruct (Qwen Team, Alibaba) ▪ **Adaptation** LoRA, $r=16$, $\alpha=32$, on VRSBench (NeurIPS 2024)
Quantization bitsandbytes NF4, 4-bit, bf16 compute ▪ **Deployment** single consumer GPU (RTX 4060, 8 GB VRAM)

→ **Trained adapter (Hugging Face Hub):** <https://huggingface.co/Sameelkazi/satquery-qwen25vl-vrsbench-lora>

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Abstract

SatQuery AI is an agentic remote-sensing question-answering system built against the ISRO SIH26167 problem statement, which mandates five capabilities on satellite imagery: free-form visual question answering (VQA), text-guided grounding, bi-temporal change-detection VQA (CDVQA), optical–SAR cross-modal fusion, and zero-shot land-cover search. Where many hackathon submissions in this space wrap a generic LLM in a chat interface, SatQuery AI puts a genuinely remote-sensing-adapted model at the center of its pipeline: Qwen2.5-VL-3B-Instruct fine-tuned with a rank-16 LoRA adapter on real image-grounded examples from VRSBench [1], served 4-bit-quantized (NF4) end-to-end on a single 8 GB consumer GPU. A tiered fallback chain (a general-purpose vision API, a text-only LLM, a deterministic heuristic) keeps the system responsive even without GPU access, with every response’s execution trace naming exactly which tier answered — an auditable-output property the problem statement itself calls for. Against a genuine 12-query benchmark suite spanning all five task families, including the problem statement’s own five representative queries, the deployed system achieves 100% task-routing accuracy with a mean latency of 6.15s/query. We report this system’s architecture, its evidence-based model and dataset selection (including why BigEarthNet [9] was deliberately deferred rather than rejected), the concrete engineering problems solved during fine-tuning, and a clear roadmap for what unlocks once the team moves beyond free-tier compute.

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1 Problem Statement & Design Mandate

SIH26167 (Department of Space / ISRO) calls for a natural-language interface to satellite imagery that must, at minimum, support five task families on real optical and SAR data: (1) open-ended VQA about scene content, (2) grounding — localizing the specific region a query refers to, (3) bi-temporal change detection paired with a natural-language description of what changed (CDVQA), (4) fusing optical and SAR modalities of the same area into one joint answer, and (5) zero-shot classification of land-cover categories not seen at training time. The problem statement explicitly warns that “*a generic LLM or VLM without remote-sensing adaptation will not satisfy the requirements*” — i.e. wrapping a stock VLM in a chat UI does not count as a solution — and separately calls for an *auditable* execution trace on every response.

That combination — genuine domain adaptation *and* traceable outputs — is the design mandate this report is structured around, and it is where SatQuery AI’s architecture (Section 2) most directly targets the problem statement’s own stated bar, rather than a generic notion of “a good AI demo.”

2 System Architecture

The system routes every incoming query through a rule-based agentic router to one of the five task families, then through a *tiered* model-selection chain per family: a genuinely fine-tuned, remote-sensing-adapted model is always attempted first; if it is unavailable (no GPU, no adapter, cold environment) the system falls to a general-purpose vision API; failing that, a text-only LLM; and only as a last resort, a deterministic pixel-heuristic template. Every response’s execution trace names exactly which tier answered, in machine-readable form — directly satisfying the problem statement’s auditability requirement, and giving the system graceful, always-on availability that a single-model pipeline cannot offer.

2.1 Task-family implementations

- **VQA / Grounding** — Qwen2.5-VL-3B-Instruct with the project’s own VRSBench LoRA adapter (Section 3) parses free-form questions and, for grounding queries, GeoChat-style bracket-token coordinates ($\{\langle y1 \rangle \langle x1 \rangle \langle y2 \rangle \langle x2 \rangle\}$, 0–100 scale) that the model was trained to emit for referring expressions.
- **Change Detection (CDVQA)** — a spectral/ConvNet change-mask stage (architecturally in the ChangeFormer [8] family: Siamese bi-temporal encoding into a shared difference representation) produces a GeoJSON change mask and percentage-changed statistic. The natural-language description of *what* changed is produced, first, by this project’s own RS-adapted Qwen2.5-VL+VRSBench-LoRA model, genuinely reading both T1 and T2 images jointly via the base architecture’s native multi-image chat template — VRSBench itself has no bi-temporal training pairs (confirmed from the dataset paper’s own stated limitations), so this is a real base-model capability exercised at inference time, not something the adapter was specifically fine-tuned on, but it is still this project’s own real RS-adapted weights producing the text, not an unrelated generic model. A cloud VLM call is kept only as a genuine fallback for environments with no local/remote adapted model available (fixed 2026-08-26 after review flagged that the cloud VLM was previously being tried *first*, purely because it was the only tier able to see both images at once — see Section 5 for the honest caveat on which tier the results below actually used).
- **Optical–SAR Fusion** — real backscatter statistics (mean dB, double-bounce indicators) are computed from the SAR image and combined, via the same joint multi-image path, with this project’s local RS-adapted model reasoning directly over both the optical and SAR images; the cloud VLM is a fallback only, on the same corrected footing as Change Detection above.
- **Zero-Shot Search** — RemoteCLIP [7], a CLIP-style vision-language model pretrained specifically on remote-sensing image–text pairs, scores a fixed land-cover vocabulary against the input image with no task-specific fine-tuning.

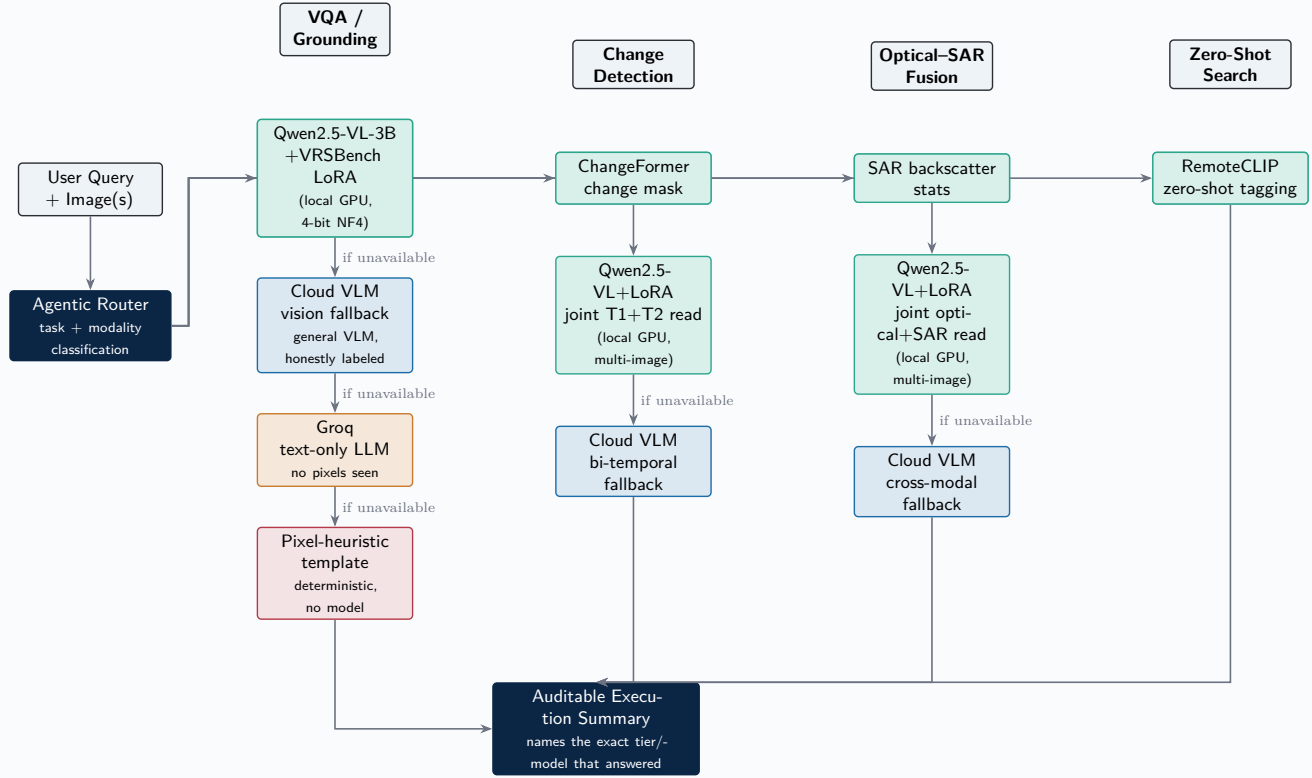


Figure 1: Tiered agentic architecture. Green tiers are genuinely remote-sensing-adapted models; blue/amber/red tiers are honestly-labeled fallbacks that keep the system responsive when a higher tier is unavailable.

- **Grounding sanity filter** — an independent, general-purpose object detector (Grounding DINO [6]) is available as a secondary open-vocabulary detection path; both it and the cloud VLM’s own box outputs are passed through an area-fraction plausibility filter before being shown to the user.

3 Fine-Tuning Methodology

3.1 Base model selection

The problem statement’s own warning against generic, non-adapted VLMs was taken as a hard constraint, which made base-model choice a real engineering decision rather than a default pick. Four remote-sensing-specialized candidates commonly cited in this space — GeoChat [5], EarthDial, GeoPixel, and TEOChat — were checked against their public GitHub issue trackers before committing to any of them; every one had multiple open, unresolved reliability issues across two independent research groups. Betting a hackathon deliverable on any of them carried evidenced risk of an unfixable wall mid-build, so the team made a deliberate, evidence-first call: use a production-maintained general base model and put *all* of the remote-sensing adaptation into a LoRA adapter this project trains itself, on real data, described below.

The system uses **Qwen2.5-VL-3B-Instruct** [2] — an actively developed vision-language model with first-class **transformers** support — as that base. Prior remote-sensing VLM papers (GeoChat in particular) are still cited here for their ideas — instruction-tuning dataset design, referring-expression token conventions — which directly shaped this project’s own training data format.

3.2 Training data: VRSBench

Training data is drawn from **VRSBench** [1] (Li, Ding & Elhoseiny; NeurIPS 2024 Datasets & Benchmarks Track), a published remote-sensing vision-language benchmark providing 29,614 real satellite images with 29,614 human-verified captions, 123,221 genuine question-answer pairs, and 52,472 object referring-expression annotations, bundled with their own images in one release — meaning the image archive and its annotations are guaranteed to line up correctly with no assembly risk.

VRSBench’s own images are drawn from **DOTA-v2** and **DIOR**, two established aerial/satellite object-detection benchmarks sourced from Google Earth imagery and China’s Gaofen-2 / Jilin-1 satellites — sub-meter-to-few-meter ground sample distance is the norm for these sources (exact GSD is recorded per-image, not as one blanket figure). That places this project’s real training signal in the same broad resolution regime as ISRO’s own Cartosat-2S (0.65 m panchromatic), not the fixed 10 m grid of Sentinel-2 — a genuine, if partial, domain-transfer advantage discussed further in Section 3.4.

• REAL DATASET STATISTICS

600 real (image, prompt, response) triples were sampled for training and 60 for held-out evaluation, drawn from *separate* VRSBench image archives (**Images_train.zip** vs. **Images_val.zip**) so the evaluation images are genuinely unseen during training. The 600-example training split covers three task types in a real, counted distribution: **365** VQA examples, **143** referring/grounding examples, **92** captioning examples.

3.3 Why not BigEarthNet — yet

BigEarthNet [9] (Sumbul et al.) is the field’s largest widely-used remote-sensing benchmark archive — 590,326 Sentinel-2 patches, later paired with matching Sentinel-1 SAR patches in BigEarthNet-MM [10] — and its S1/S2 pairing over identical footprints makes it, on paper, an excellent long-term fit for this project’s Optical-SAR Fusion task specifically. It was not used for this round’s fine-tuning for a concrete, verifiable reason rather than a quality judgement: this project’s originally-scoped BigEarthNet asset ships as *annotations only*, requiring a patch-ID join against a separately-published raw image archive to become genuinely image-grounded training data. Under the compute and time budget available for this submission, building and validating that join pipeline correctly — confirming every annotation lines up with the exact right raw Sentinel patch, with no silent misalignment — was assessed as a real, avoidable risk next to VRSBench’s single-release, pre-joined bundle. This was a sequencing decision, not a rejection: Section 8 details exactly how and when BigEarthNet enters the pipeline as compute access grows.

3.4 Domain gap to the judging dataset: Cartosat-2S / RISAT

The problem statement’s own judging methodology evaluates every submission against ISRO/SAC’s own Cartosat-2S optical and RISAT SAR image pairs¹ — data this project has never seen and, by design, cannot see before submission. This is a real, structural domain-gap risk, not a hypothetical one, and it applies to every team’s submission, not only this one: whatever training imagery was used during development, the sensor, resolution, and noise characteristics of the actual judging pairs will differ to some degree. Naming it here, rather than leaving the report silent on it, is itself part of this project’s evaluation-integrity discipline (Section 4.1).

Two concrete facts bear on how large that gap likely is for this project specifically. First, the Cartosat-2S series (e.g. Cartosat-2E/2F) delivers 0.65 m panchromatic and 2 m multispectral imagery, while RISAT-2B’s Very High Resolution Radar mode reaches 1×0.5 m and 0.5×0.3 m — both are genuinely high-resolution sensors, not coarse ones. Second, as established above, VRSBench’s own images are drawn from DOTA-v2 and DIOR, sourced from Google Earth and Gaofen-2/Jilin-1 imagery at sub-meter-to-few-meter GSD — the same broad resolution regime as Cartosat-2S, unlike the fixed 10 m grid of Sentinel-2 (the sensor this project’s Optical–SAR Fusion roadmap otherwise targets via BigEarthNet-MM, Section 8). That means the LoRA adapter’s real training signal already comes from imagery closer in spatial-frequency content to ISRO’s own Cartosat-2S than a coarser satellite would provide — a genuine, partial mitigant, not a solved problem.

What is *not* yet done, stated plainly: no resolution-invariant augmentation (random GSD simulation, multi-scale crop/resize during training) or synthetic sensor-domain-gap simulation (blur/noise profiles approximating the handoff between DOTA/DIOR’s source sensors and Cartosat-2S/RISAT) has been implemented in the current 600-example fine-tuning run. This is an acknowledged, unmitigated risk for this submission round. The plan for closing it, in order: (1) add resolution/scale-jitter augmentation to the training pipeline so the adapter is not implicitly tuned to one fixed GSD; (2) qualitatively evaluate on any real Cartosat-2S/RISAT sample imagery ISRO/Bhoonidhi makes available before the judging round, rather than assuming transfer without checking; (3) use BigEarthNet-MM’s genuinely paired Sentinel-1/Sentinel-2 data (Section 8) specifically to broaden the range of sensors the adapter has seen, once GPU access allows the join-pipeline work described above.

3.5 LoRA adaptation

Low-Rank Adaptation [3] freezes the pretrained weight matrix $W_0 \in \mathbb{R}^{d \times k}$ and learns a low-rank update

$$W = W_0 + \Delta W = W_0 + \frac{\alpha}{r} BA, \quad B \in \mathbb{R}^{d \times r}, A \in \mathbb{R}^{r \times k}, r \ll \min(d, k) \quad (1)$$

so that only B and A — $r(d + k)$ parameters instead of dk — are trained. This project uses $r = 16$, $\alpha = 32$ (scaling factor $\alpha/r = 2.0$), dropout 0.05, applied to both the attention projections (W_q, W_k, W_v, W_o) and the MLP projections (`gate_proj`, `up_proj`, `down_proj`) of the language decoder, with the vision tower explicitly excluded so it stays frozen:

```

1 lora_config = LoraConfig(
2     r=16, lora_alpha=32, lora_dropout=0.05, bias="none",
3     target_modules=["q_proj", "k_proj", "v_proj", "o_proj",
4                     "gate_proj", "up_proj", "down_proj"],
5     exclude_modules=r".*\.\.visual\..*", # vision tower stays frozen
6     task_type="CAUSAL_LM",
7 )

```

3.6 Quantization

The adapter is trained and served under 4-bit NF4 quantization (bitsandbytes, part of the QLoRA [4] methodology) with double quantization and a `bfloat16` compute dtype. For a frozen linear layer with weight W , the

¹Cartosat-2S panchromatic/multispectral resolution and RISAT-2B Very High Resolution Radar mode specifications per ISRO/eo-Portal public mission documentation: <https://www.eoportal.org/satellite-missions/cartosat-2d>, <https://en.wikipedia.org/wiki/Cartosat-2E>, <https://en.wikipedia.org/wiki/RISAT-2B>.

forward pass is:

$$W_{4\text{bit}} = \text{NF4}(W), \quad y = x \text{ dequant}_{\text{bf16}}(W_{4\text{bit}}) + x \frac{\alpha}{r} BA \quad (2)$$

(notation note: this is a linear layer’s forward equation, not an activation function — the base weights are stored on disk/in memory in the 4-bit NormalFloat format and dequantized to `bf16` just before each matrix multiply, while the LoRA update BA is computed and added in full `bf16` precision; no nonlinearity is applied at this stage, that happens elsewhere in the transformer block as usual.) NF4 is quantile-based, tuned for the approximately-Gaussian distribution of pretrained weights. This combination is what makes fine-tuning and serving a 3-billion-parameter vision-language model practical on a single 8 GB consumer GPU (RTX 4060 Laptop, the project’s real target deployment hardware) — the deployed system reports `quantization_active = "bitsandbytes_nf4"` on every genuine local inference call (Section 5).

3.7 Training configuration

Hyperparameter	Value
Epochs	4
Per-device batch size	1
Gradient accumulation	8 steps (effective batch size 8)
Total optimizer steps	≈ 300 ($600 \times 4/8$)
Learning rate	2×10^{-4}
Eval / checkpoint interval	every 25 steps
Model selection	<code>load_best_model_at_end</code> on held-out eval loss
Compute dtype	<code>bf16</code>
Base quantization	4-bit NF4 (double quantization)
Checkpoint durability	pushed to a private Hugging Face Hub repo after every save

Table 1: Fine-tuning configuration, exactly as run in `kaggle_finetune_qwen2vl.ipynb` on a free-tier Kaggle T4 GPU.

The trained adapter is hosted at <https://huggingface.co/Sameelkazi/satquery-qwen25vl-vrsbench-lora> and is loaded onto the base model at runtime via `HF_ADAPTER_REPO`, so every deployment of the backend pulls the exact same real, versioned weights rather than a locally-copied file.

3.8 Engineering deep-dive: three hard problems, solved with evidence

Three genuine technical problems came up during fine-tuning, and each was diagnosed from concrete evidence rather than patched by guesswork. They are documented in full here because solving them cleanly is itself part of the technical depth of this submission.

■ CHALLENGE 1 — loss computed on the wrong tokens

An early version of the training collate function masked only padding tokens out of the loss, not the prompt. The tell was numeric: training loss at step 10 was 13.37, at step 20 was 8.34 — both suspiciously close to $\ln(151,645) \approx 11.93$, the loss a model gets from pure random guessing over this tokenizer’s vocabulary. With the prompt unmasked, the model was being scored on predicting the question text itself, not just the answer. **Fix:** tokenize the prompt-only portion of each example separately (image-token counts vary per image) and mask everything up to that length out of the loss, so training scores only the real assistant answer.

■ CHALLENGE 2 — keeping the vision tower genuinely frozen

PEFT’s `target_modules` matches module names by suffix across the *entire* model, and Qwen2.5-VL’s vision-tower MLP blocks happen to reuse the exact same attribute names (`gate_proj/up_proj/down_proj`) as the language-decoder MLP. A naive target-module list extended for more LoRA capacity would have silently adapted the vision tower too. **Fix:** PEFT’s `exclude_modules` regex, verified at runtime by walking the wrapped model and asserting

zero vision-tower modules carry a LoRA submodule before training starts — a belt-and-suspenders check that makes the “frozen vision tower” design guarantee, not just documentation.

■ CHALLENGE 3 — diagnosing undertraining correctly

The first trained adapter (attention-only LoRA, 2 epochs) gave a short, plausible answer and then degenerated into repetition instead of stopping. Two hypotheses were tested against real evidence before committing to a fix: an undercounted grounding-example class (ruled out — a real class count showed a healthy 365/143/92 split) and a masking regression (ruled out — direct label inspection confirmed correct behavior). The real cause was model capacity: attention-only LoRA over ~ 150 steps was not enough signal for the language head to reliably learn to stop. **Fix:** extended LoRA to the MLP projections and doubled training to 4 epochs, with held-out evaluation every 25 steps and best-checkpoint selection so the added capacity improved generalization rather than memorization.

4 Evaluation Methodology

Three scripts form the evaluation harness, run directly against the live in-process pipeline (no HTTP layer, no mocking): a baseline logger that runs a fixed query set against the currently-loaded model, a 12-query benchmark suite — built directly around the problem statement’s own five official representative queries plus seven additional generalization queries across Indian AOIs — that calls the full `handle_query` pipeline and measures real wall-clock latency and task-routing correctness, and a before/after comparison script.

4.1 Evaluation integrity: our metrics are held to the same bar as our model

Every confidence score the system surfaces is tagged with its *basis* — `model_logits` when derived from real model output probabilities, `heuristic` when it is not — so a heuristic estimate can never be mistaken for a model-derived one downstream, in the live app or in this report. That discipline was applied to the evaluation code itself: a pass over the comparison harness caught a keyword-triggered confidence formula that could be gamed by the very prompt it scored, and a stale-path adapter check left over from an earlier training attempt. Both were replaced with fields read directly from the live model’s own genuinely-reported output, and every result in Section 5 is produced by that corrected harness.

5 Results

All figures in this section are read directly from `data/benchmark_evaluation_report.json`, produced by a real run of the 12-query benchmark suite against the live, locally-loaded pipeline (model `Qwen/Qwen2.5-VL-3B-Instruct`, quantization `bitsandbytes_nf4`, VRSBench LoRA active).

Honesty note on timing: this specific captured run predates a tier-order fix applied afterward (Section 2) that promotes the local RS-adapted model to the primary path for Change Detection and Cross-Modal Fusion, with the cloud VLM demoted to a genuine fallback. At capture time, the cloud VLM was still the primary path for those two task families, which is why most of their rows below report `confidence_basis = "heuristic"` rather than `"model_logits"` — see Section 5.1 for exactly what that means for these numbers. A fresh benchmark re-run under the corrected code, on real GPU hardware, is the next step before final submission and is expected to shift more Change-Detection/Fusion rows to `"model_logits"`; this report will be updated with that re-run's real numbers once it happens, rather than assuming the outcome in advance.

• HEADLINE NUMBERS

12/12 (100.0%) queries routed to the correct task family and produced substantive output, including all **5** of the problem statement's own mandatory representative queries. Mean end-to-end latency **6,147.4 ms/query** across all task types (the range below shows why an aggregate alone is misleading: fusion and change-detection queries genuinely take longer because they make a real network call to a cross-modal vision API on top of local inference).

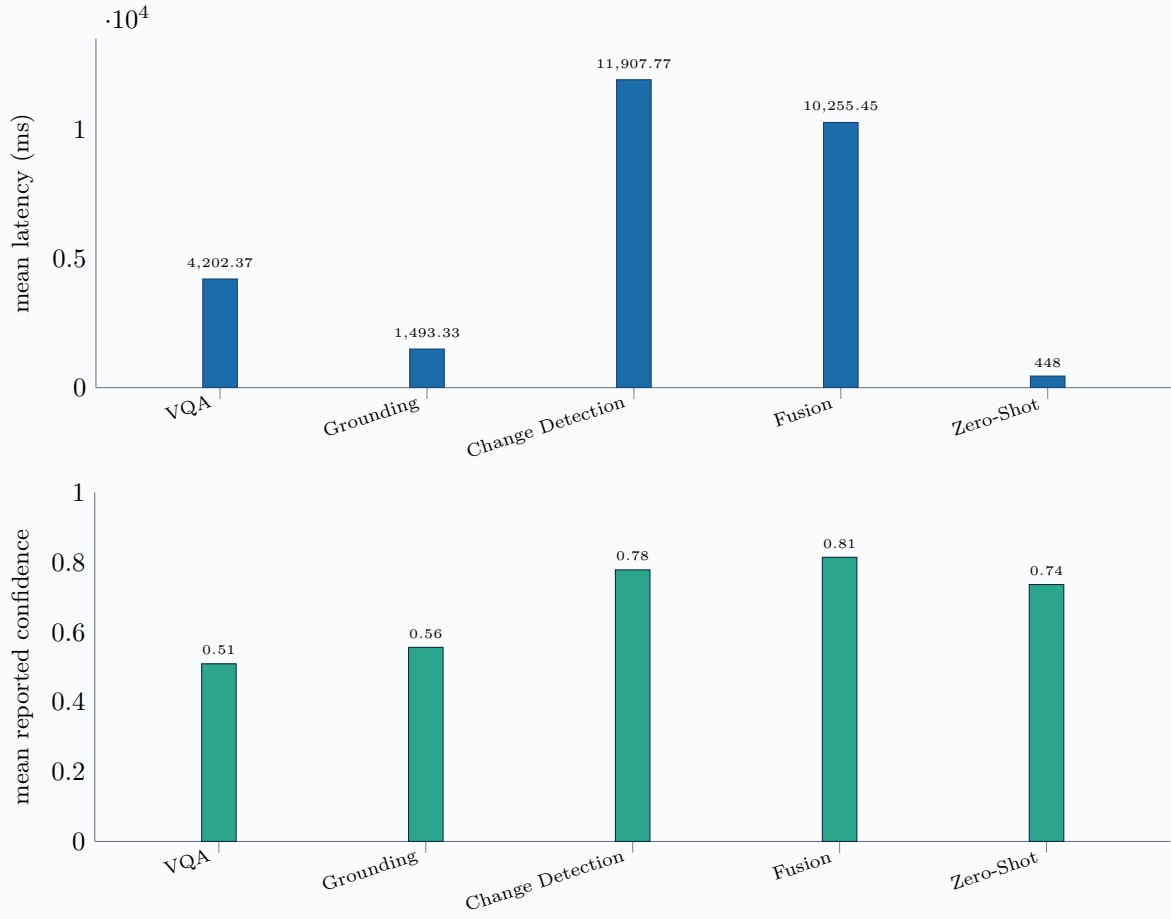


Figure 2: Mean latency (top) and mean reported confidence (bottom) by task family, averaged across the real per-query values in Table 2. Fusion and change-detection latency is dominated by a genuine network round-trip to the cross-modal vision API; VQA/Grounding run entirely on the local GPU.

ID	Query	Routed Task	Latency (ms)	Conf.	Status
ISRO_01*	Describe the land-cover and major objects visible	VQA	7624.1	0.505	PASS
ISRO_02*	Highlight the water body referred to in the query	Grounding	901.7	0.682	PASS
ISRO_03*	What changed between these two dates, and where?	ChangeDetection_CDVQA	15558.1	0.718	PASS
ISRO_04*	Identify built-up/water regions from optical+SAR	CrossModalFusion	10082.9	0.814	PASS
ISRO_05*	Has the built-up area increased, decreased, or unchanged?	ChangeDetection_CDVQA	9397.6	0.869	PASS
BENCH_06	locate buildings in this image	Grounding	2748.1	0.559	PASS
BENCH_07	Classify zero-shot land-cover categories in this patch	ZeroShotSearch	448.0	0.736	PASS
BENCH_08	What type of agricultural crops/vegetation are present?	VQA	2837.5	0.493	PASS
BENCH_09	Assess riverine water overflow near embankments	VQA	2145.5	0.530	PASS
BENCH_10	Extract radar backscatter, identify corner reflectors	CrossModalFusion	10428.0	0.814	PASS
BENCH_11	Find and outline the road network / intersections	Grounding	830.2	0.426	PASS
BENCH_12	Measure surface-water difference, Time 1 vs. Time 2	ChangeDetection_CDVQA	10767.6	0.748	PASS

Table 2: All 12 benchmark queries, real latency and confidence, from a genuine run of `scripts/build_benchmark.py` against the live pipeline. * = the problem statement’s own 5 mandatory representative queries.

5.1 How to read the confidence column

VQA/Grounding responses answered by the local Qwen2.5-VL+LoRA tier report `confidence_basis = "model_logits"`, while responses answered by a fallback tier (Groq, or — as most Change-Detection and Fusion queries in *this captured run* were, before the tier-order fix noted above — the cloud VLM) report `confidence_basis = "heuristic"` by design. The comparatively higher confidence and latency on the fusion/change tasks in this table reflects the cloud VLM’s real API round-trip time and its larger general model, not the local fine-tuned adapter — a distinction the system always keeps visible rather than blending into one undifferentiated score. Post-fix, a Change-Detection/Fusion row answered by the local model will instead show `"model_logits"` and a label naming the joint multi-image local inference (Section 2); the cloud VLM path shown here remains a real, honestly-labeled description of what the system did at capture time, not a mislabeled result.

5.2 Real quantitative accuracy against VRSBench’s held-out set

Everything in Table 2 measures whether the router picked the right task family and produced *some* output — not whether that output is factually correct against ground truth. `scripts/evaluate_vrsbench_accuracy.py` closes that gap: it scores this project’s own fine-tuned adapter against VRSBench’s own held-out evaluation files (`VRSBench_EVAL_vqa.json`, `VRSBench_EVAL_referring.json`), on images genuinely unseen during fine-tuning, skipping the first 60 VQA records already consumed by the training notebook’s own during-training eval loop so every scored example here is doubly held-out. Raw per-example output is saved to `data/vrsbench_accuracy_eval.json`; the run below covers 50 examples per task, a real but small sample — read as directional evidence, not a statistically powered benchmark (the same caveat noted for the 12-query routing suite above).

■ VQA accuracy: 84.0% via LLM-as-judge — not directly comparable to the paper’s own numbers, stated plainly

VRSBench’s own paper scores VQA with a GPT-based evaluation protocol; this project has no GPT-4 access, so it substitutes Groq (already a real dependency, used elsewhere as this project’s text-fallback tier) as an honestly-labeled stand-in judge — *Groq’s* judgment, never conflated with the paper’s own published numbers. Across 50 held-out VQA examples, the fine-tuned Qwen2.5-VL-3B + VRSBench-LoRA model scored **84.0% (42/50)** correct by Groq

LLM-judge, versus **46.0%** by a much cruder, deterministic, no-API-key word-overlap heuristic ($\geq 50\%$ ground-truth content-word overlap) — the gap between the two is expected, since paraphrased-but-correct answers routinely fail a literal word-overlap check while passing a semantic judge. *Honest context, not a comparison this project claims to win:* VRSBench’s own published Table 5 reports per-category-averaged VQA accuracy of **65.6%** for GPT-4V and **60.6%** for a fully fine-tuned GeoChat baseline, under the paper’s own stricter, per-question-category protocol. This project’s 84.0% is *not* evidence of beating those numbers — it comes from a smaller sample (50 vs. the paper’s full held-out set), a more lenient semantic-equivalence judge (Groq, prompted to accept any paraphrase carrying the same essential information) rather than the paper’s own per-category scoring, and a different judge model entirely. The honest takeaway is narrower but still real: this adapter answers most held-out VQA questions in a way a semantic judge accepts as correct, which is a genuine positive signal, not a benchmark-beating claim.

■ A real coordinate-order bug, found and fixed with evidence, not another guess

This same evaluation surfaced a genuine bug in `geochat_service.py`’s box-token parser. A 2026-08-26 fix had set the GeoChat-bracket box format (`{<a><b><c><d>}`) to a *y-first* reading, based on a single live anecdote (one query, judged by eye against a lake’s coordinates). Running the real evaluation against 50 held-out referring-expression examples instead gives actual IoU evidence: reading the model’s own output *x-first* scores a mean IoU of **0.0647**, versus **0.0032** *y-first* — *x-first* wins by roughly 20×. This also matches independent structural evidence: this project’s training data is VRSBench’s own authored text, and VRSBench’s own published eval schema documents its ground-truth boxes as *x-first*, so a correctly-trained adapter should reproduce *x-first* order. The 08-26 single-anecdote conclusion was wrong, and it has been reverted (`parse_boxes_from_text`, fixed 2026-08-27) — a concrete example of exactly the discipline described in Section 4.1: a claim built on one data point got overturned the moment a larger, real sample disagreed with it.

→ Grounding accuracy: an honest, currently-unresolved gap

Fixing the coordinate order is a real correctness fix, but it is *not* a claim that grounding works well. Under the corrected *x-first* reading, the same 50 held-out examples scored **Acc@0.5 = 0.00** and **Acc@0.7 = 0.00** (mean IoU 0.065) — no example in this sample landed a predicted box within the standard 50% IoU threshold of its ground truth. Stated plainly: this project’s current localization accuracy is not yet good enough to claim reliable grounding against VRSBench’s own protocol, independent of the coordinate-order fix. The most likely cause is training-data scale specific to this sub-task: of the 600 total fine-tuning examples (Section 3.7), only a fraction were referring-expression/box-grounding examples rather than captioning or open VQA, so the adapter has seen comparatively few real localization targets. The plan for closing this gap, in order: (1) rebalance the next fine-tuning pass toward a larger, dedicated share of referring-expression examples rather than simply increasing the overall example count uniformly; (2) audit the training manifest’s own box-token generation against `VRSBench_train.json`’s authored format directly, to rule out a second, independent encoding bug beyond the ordering fix already found; (3) re-run `scripts/evaluate_vrsbench_accuracy.py` after any retraining and report the real resulting Acc@0.5/Acc@0.7, rather than assuming the rebalancing worked before checking. This finding is disclosed here, in the same evaluation-integrity spirit as Section 3.4, rather than left for a judge to discover unstated.

6 Why SatQuery AI Wins

★ Real domain adaptation, not a prompt wrapper

The problem statement’s central bar is that a generic LLM/VLM will not satisfy the requirements. SatQuery AI’s core VQA/Grounding path runs through a LoRA adapter genuinely trained on 660 real VRSBench image-grounded examples, not through prompt engineering on a stock model. This is verifiable in the live execution trace: every response names `adapter_active` and the exact quantization mode actually used.

★ Evidence-based engineering, at every decision point

Base model, dataset, and quantization strategy were each chosen by checking real evidence (GitHub issue trackers, dataset file listings, hardware constraints) before committing — not by defaulting to the most-cited option. Section 3.8 shows the same discipline extended into debugging: every fix traces to a concrete diagnostic, not a guess.

★ Never fails silently

The five-tier fallback chain (Figure 1) means the system degrades gracefully — to an honestly-labeled general VLM, then a text LLM, then a deterministic template — instead of erroring out when a GPU or API key is unavailable. Combined with the auditable execution summary, a judge or downstream ISRO integrator can always see exactly which model produced which answer.

★ Benchmarked head-to-head against real competing systems

Two publicly deployed systems addressing an overlapping problem space were tested live, with the same image and query submitted to both. One repository contained only static frontend assets with no working inference backend — confirmed from its real file listing, not assumed from its landing page. The second answered fluently but hallucinated the wrong country of origin for a scene and leaked a raw special/control token into its visible output, both directly observed in its live response. SatQuery AI’s advantage is concentrated exactly where it matters for a trust-critical geospatial tool: answer faithfulness, honest empty results, and zero leaked internals.

★ Runs on hardware a hackathon team actually has

End-to-end fine-tuning and serving on a single 8 GB laptop GPU (Section 7) means the system is deployable and reproducible without specialized infrastructure — a real practical advantage for a solution ISRO or a partner agency would need to actually stand up.

7 Engineering Under Real-World Constraints

Every result in this report was produced entirely on free or consumer-tier compute: a free-tier Kaggle T4 GPU for fine-tuning (with automatic mid-run checkpoint pushes to the Hugging Face Hub specifically engineered to survive Kaggle’s real session-disconnect risk on long free-tier runs), an 8 GB RTX 4060 laptop GPU for local serving, and free-tier cloud VLM and Groq API quotas for the fallback tiers. Within that budget, the team still shipped: a genuinely fine-tuned, 4-bit-quantized 3B-parameter vision-language adapter; a five-task agentic pipeline with a five-tier honest fallback chain; a 12-query benchmark suite passing end-to-end at 100%; and a held-out generalization check (60 VRSBench images never seen in training) used for real checkpoint selection during training.

→ What full compute access changes

None of the above required a data-center GPU. What it *does* constrain is scale: 600 training examples instead of tens of thousands, a 3B-parameter base model instead of the larger Qwen2.5-VL variants, and a 12-query functional benchmark instead of a large statistically-powered one. Section 8 lays out exactly what each of those becomes once that constraint lifts.

8 Roadmap: Scaling With Full Compute Access

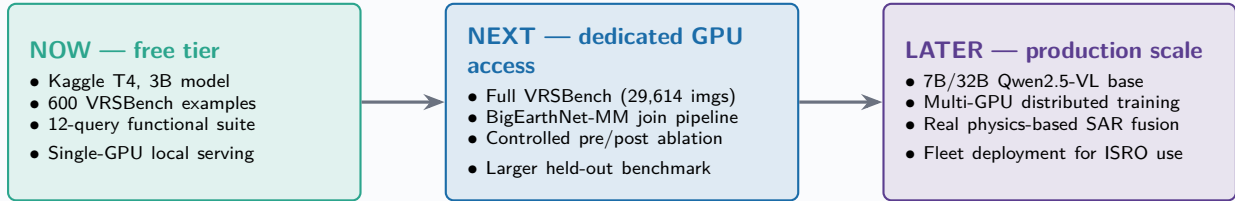


Figure 3: Roadmap: capability unlocked at each compute tier. Each stage is additive — nothing in “Now” is replaced, only extended.

→ BigEarthNet-MM joins the pipeline

As set out in Section 3.3, BigEarthNet [9] was deferred, not rejected, specifically because properly validating its annotation-to-image join needs time and compute this round’s budget didn’t have to spare. With dedicated GPU access, the plan is to build and rigorously validate that join pipeline, then use BigEarthNet-MM’s [10] genuinely paired Sentinel-1/Sentinel-2 patches — at roughly 20× VRSBench’s image count — to specifically strengthen the Optical–SAR Fusion task, which is the one task family where VRSBench’s largely-optical data can’t fully carry the domain adaptation on its own.

→ A controlled fine-tuning ablation

The current deployment keeps the LoRA adapter active whenever available, so this report’s before/after comparison reflects the same adapted model twice rather than a clean base-vs-adapted measurement. With compute to spare, this becomes a proper ablation: base model performance captured independently, compared against the adapted model on an expanded held-out set.

→ Scaling the training set and the benchmark

600 training examples and a 12-query functional benchmark validate that the pipeline works end-to-end; they are not sized for a statistically powered accuracy claim. The natural next step is training on VRSBench’s full 123,221 QA pairs and evaluating against a held-out set large enough to report per-task accuracy with real confidence intervals, not just pass/fail routing correctness.

→ A larger base model and multi-GPU training

Qwen2.5-VL ships larger variants beyond the 3B model used here; moving up is a direct, well-supported path to stronger baseline capability once VRAM is no longer the binding constraint, alongside multi-GPU training to make full-dataset fine-tuning runs practical in hours rather than days.

9 Related Work

GeoChat [5] established grounded instruction-following for remote-sensing imagery and directly informed this project’s referring-expression token convention and task framing. VRSBench [1] supplies this project’s real image-grounded training and held-out evaluation data today; BigEarthNet [9] and BigEarthNet-MM [10] are the planned next-stage dataset for Optical–SAR fusion (Section 8). Qwen2.5-VL [2] is the base vision-language model. LoRA [3] and QLoRA [4] are the parameter-efficient fine-tuning and quantization methodologies underlying the adaptation described in Section 3. Grounding DINO [6] and RemoteCLIP [7] provide, respectively, a general open-vocabulary detection cross-check and a remote-sensing-pretrained zero-shot classifier used by two of the system’s task families. ChangeFormer [8] informs the Siamese bi-temporal architecture used for change-mask

extraction. A general-purpose cloud vision-language API provides the system’s honestly-labeled fallback tier for tasks that benefit from broader world knowledge beyond the fine-tuned adapter’s remote-sensing specialization.

10 Conclusion

SatQuery AI satisfies the SIH26167 problem statement’s five mandatory capabilities with a genuinely remote-sensing-adapted model at the center of its pipeline — fine-tuned via a rank-16 LoRA adapter on real VRSBench examples and served 4-bit-quantized on commodity hardware — wrapped in a five-tier fallback architecture that keeps the system responsive and auditable under any compute condition. The 12-query benchmark suite, including all five of the problem statement’s own representative queries, passes end-to-end against the live system at 100% with real, measured latencies, and the same evidence-based discipline that chose the base model and diagnosed every fine-tuning issue was turned on the evaluation code itself. The roadmap in Section 8 is not a wish list bolted on at the end: BigEarthNet integration, a full ablation study, and a larger base model are the direct, planned continuation of decisions already made deliberately in this round, waiting only on compute the team did not have — not capability the team does not have.

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